Section 4.4: Part B: Probability and the Standard Distribution

Finite Mathematics MTH 107 Fall 2014

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Mars Hill University

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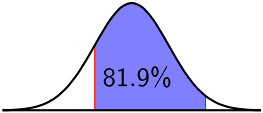
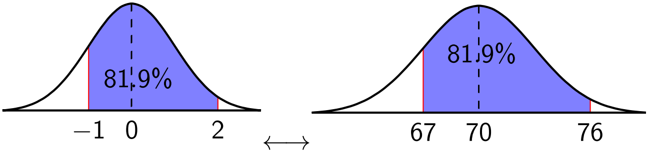
Two key ideas

1. Area under the curve between two values
   = Percent of population between those values.



2. Percent of population between two values of any normally distributed population can be found by using the *standard distribution.*



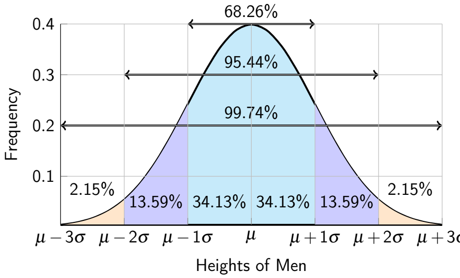
Example 4.4.1.

The *standard normal z-distribution* has a mean of 0 and standard deviation of 1.

a. What percentage of the standard distribution lies between $z = -1$ and $z = 2$?

b. What is the probability that a randomly selected data point has a value between 1 and 2?

c. Find $p(z < 70)$.



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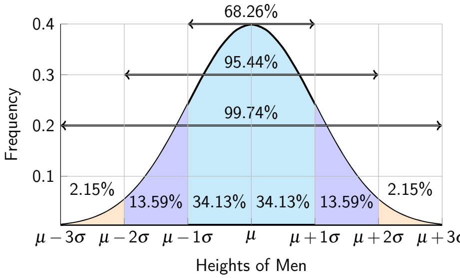

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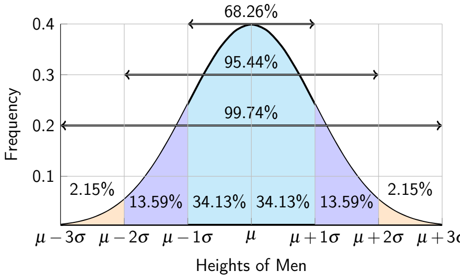

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b. 81.85%

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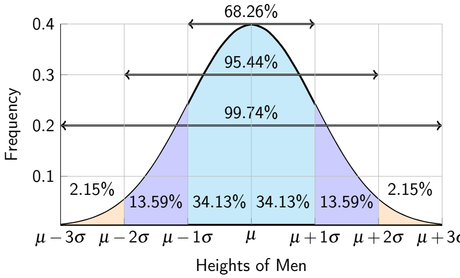


a. $68.26\% + 13.59\% = 81.85\%$
b. 81.85%
C. 50%

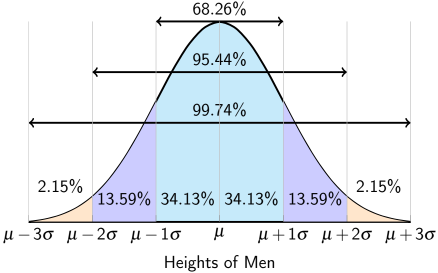
Example 4.4.2.

The mean height of males is 70 inches with a standard deviation of 3 inches.

a. What percentage of men are between 67 and 76 inches?

b. What is the probability that a randomly selected man has a height between 67 and 76 inches?

c. Find $p(x > 70)$.



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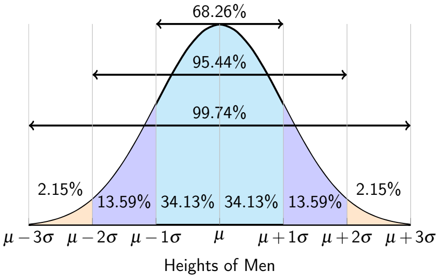

a. $68.26\% + 13.59\% = 81.85\%$

Normal Distribution: Probability

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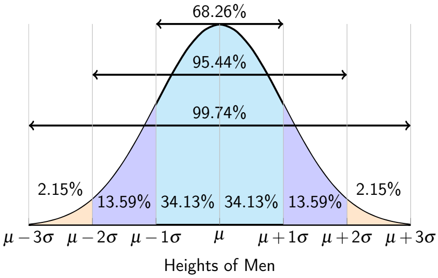

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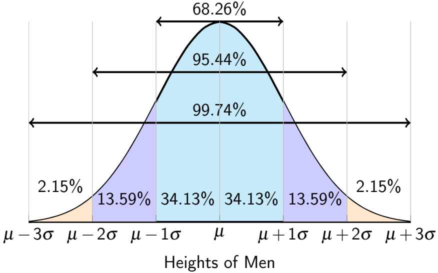


a. $68.26\% + 13.59\% = 81.85\%$
b. 81.85%
c. 50%

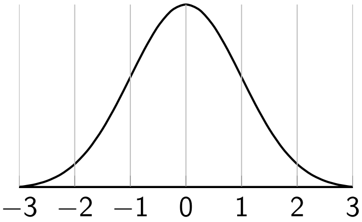
Standard Distribution: Probability

Applet: http://www.mathsisfun.com/data/standard-normal-distribution-table.html
Chart: http://www.stat.ufl.edu/ athienit/Tables/Ztable.pdf

Example 4.4.3.

Find the following probabilities.

a. $p(z < 1.65)$
b. $p(0 < z < 1.65)$
c. $p(z < -0.67)$
d. $p(-.067 < z)$
e. $p(-.67 < z < 1.65)$



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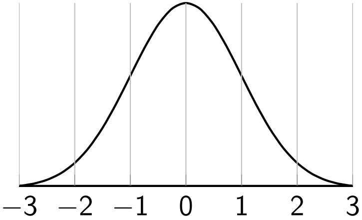

a. 95.05%

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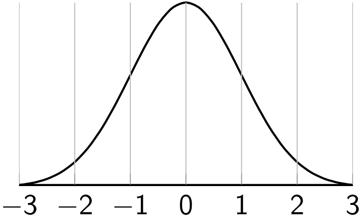

a. 95.05%
b. $95.05\% - 50\% = 45.05\%$

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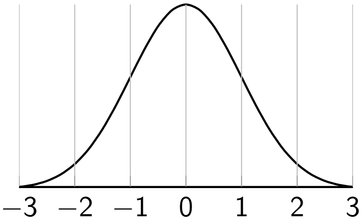

a. 95.05%
b. $95.05\% - 50\% = 45.05\%$
c. 25.14%

Standard Distribution: Probability

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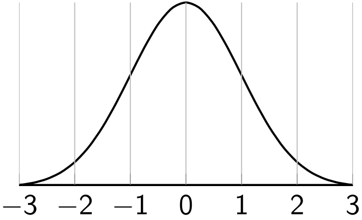

a. 95.05%
b. $95.05\% - 50\% = 45.05\%$
c. 25.14%
d. $100\% - 25.14\% = 74.86\%$

Standard Distribution: Probability

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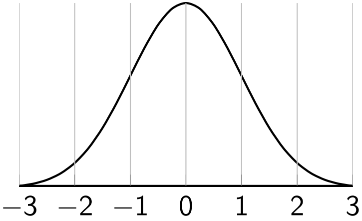

a. 95.05%
b. $95.05\% - 50\% = 45.05\%$
c. 25.14%
d. $100\% - 25.14\% = 74.86\%$
e. $24.86\% + 45.05\% = 69.91\%$

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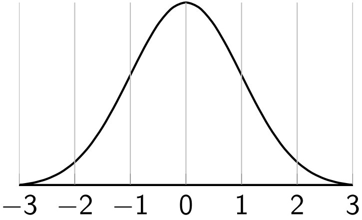
Example 4.4.4.

Find the following probabilities.

a. $p(z < 1.42)$
b. $p(0 < z < 1.42)$
c. $p(-0.87 < z < 1.66)$
d. $p(-1.03 < z < 1.66)$
e. $p(-.67 < z < 1.65)$



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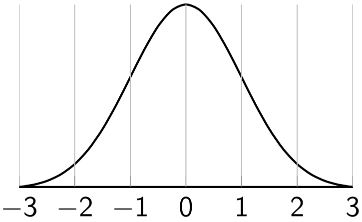
c. $p(-0.87 < z < 1.66)$
d. $p(-1.03 < z < 1.66)$
e. $p(-.67 < z < 1.65)$
f. Let $\mu = 5$ and $\sigma = 10$; $p(x < 19.2)$

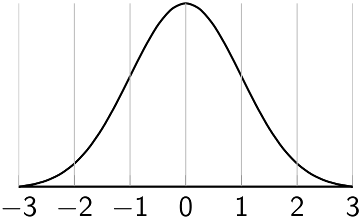


Example 4.4.5.

A population is normally distributed with a mean of 5 and a standard deviation of 10. Find the following probabilities.

a. $p(x < 19.2)$

b. $p(10 < x)$



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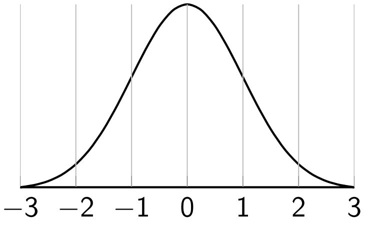

Convert a Normal to a Z

If x is in a normal distribution then it is

$$z = \frac{x - \mu}{\sigma}$$

standard deviations from the mean.